2025 SSMO Team Round Solutions

SMO Team

TR 1: There are N solutions (a, b, c) to $a + b^2 + c = 2025$, where a and b are positive integers and c is a nonnegative integer. Find the number of positive factors of N.

Answer: $\boxed{32}$

Solution: Rearrange the given equation as $a + c = 2025 - b^2$ and fix b. The possible values of a are $1, 2, \ldots, 2025 - b^2$, and the corresponding possible values of c are $2025 - b^2 - 1, 2025 - b^2 - 2, \ldots, 0$. Therefore, there are $2025 - b^2$ solutions (a, c) to $a + c = 2025 - b^2$. The possible values of b are $1, 2, \ldots, 44$, so

$$\begin{aligned} N &= \sum_{b=1}^{44} (2025 - b^2) \\ &= \sum_{b=1}^{44} 2025 - \sum_{b=1}^{44} b^2 \\ &= 44 \cdot 2025 - \frac{44 \cdot 45 \cdot 89}{6} \\ &= 59730. \end{aligned}$$

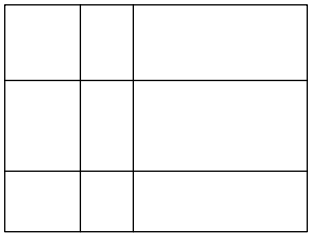
Since $59730 = 2 \cdot 3 \cdot 5 \cdot 11 \cdot 181$, the number of positive factors of N is $2^5 = \boxed{32}$.

TR 2: Nonnegative integers a and b satisfy $a! + b^4 = 16a + 41$. Find the sum of all possible values of b.

Answer: $\boxed{4}$

Solution: Rearrange the given equation as $b^4 = 16a + 41 - a!$. The left hand side is always nonnegative, but the right hand side is only nonnegative when $a \leq 5$. If $a = 0, 1, 2, 3$, then $b^4 = 40, 56, 71, 83$, none of which are fourth powers. If $a = 4$, then $b^4 = 81$, so $b = 3$, and if $a = 5$, then $b^4 = 1$, so $b = 1$. So, the sum of all possible values of b is $3 + 1 = \boxed{4}$.

TR 3: A rectangle is divided as shown into nine smaller rectangles. The areas of the five smallest rectangles are 1, 2, 3, 4, and 5. What is the largest possible area of the original rectangle?



Answer: $\boxed{64}$

Solution: Note that permuting the rows or the columns does not change the multiset of areas, and the rectangle with area 1 lies in the shortest row and the narrowest column, since it is the smallest. So, WLOG, assume that the rectangle with area 1 is in the bottom-left corner. We label the areas and the side lengths of the sub-rectangles as shown.

c	Q	X	Z
b	P	W	Y
a	1	R	S
	d	e	f

Then, we have

$$\begin{aligned} W &= abde = PR \\ X &= acde = QR \\ Y &= abdf = PS \\ Z &= acdf = QS. \end{aligned}$$

Because $W = PR$ and $PR \geq 2 \cdot 3 = 6$, we must have $W \notin \{2, 3, 4, 5\}$. By analogous reasoning, we can conclude that $X, Y, Z \notin \{2, 3, 4, 5\}$ as well. Therefore, $\{P, Q, R, S\} = \{2, 3, 4, 5\}$. We know that $P + Q + R + S = 2 + 3 + 4 + 5 = 14$. Thus, in order to maximize the area of the original rectangle, it suffices to maximize the value of $W + X + Y + Z$. We have

$$\begin{aligned} W + X + Y + Z &= PR + QR + PS + QS \\ &= (P + Q)(R + S). \end{aligned}$$

We know that $P + Q + R + S = 14$, so the expression $(P + Q)(R + S)$ is maximized when $P + Q = R + S = 7$, which is achievable with $(P, Q, R, S) = (2, 5, 3, 4)$. This yields $7^2 = 49$ as the largest possible value of $W + X + Y + Z$. Hence, the largest possible area of the original rectangle is $1 + 14 + 49 = \boxed{64}$.

TR 4: Let $P(x) = x^2 - ax + b$ be a quadratic with nonzero real coefficients. Given that $P(a)$ and $P(-b)$ are roots of $P(x)$, there exists a value of c such that $P(c)$ is constant for all possible $P(x)$. Find c.

Answer: $\boxed{1}$

Solution: Note that $P(a) = b$. Since b is always a root of $P(x)$, from Vieta's formulas, 1 is always a root of $P(x)$ as well. Therefore, we have $P(1) = 0$ over all possible $P(x)$. We now show that there is no value of c other than 1 such that $P(c)$ is constant over all possible $P(x)$. If there exist at least two such quadratics $P(x)$, we are done because any two distinct monic quadratics cannot be equal on more than one input. It is easy to check that the three quadratics $(x-1)(x+\frac{1}{2})$, $(x-1)(x-\frac{-1+\sqrt{3}}{2})$, and $(x-1)(x-\frac{-1-\sqrt{3}}{2})$ satisfy the conditions imposed on $P(x)$. (In fact, these are the only possible $P(x)$; finding them

is only a matter of showing that $P(P(a)) = 0$ implies $a = b + 1$, and then determining the possible values of b from the equation $P(P(-b)) = 0$.) Thus, the answer is $\boxed{1}$.

TR 5: Bob rolls a fair six-sided die until he rolls an even number. What is the expected sum of all the numbers he rolled?

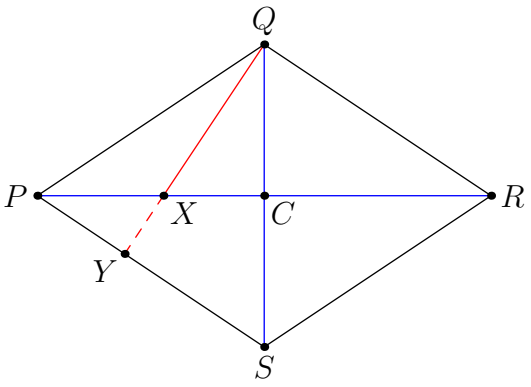
Answer: $\boxed{7}$

Solution: Let E denote the expected sum of all the numbers Bob rolls. If Bob's first roll is an odd number n_o, his expected sum is $n_o + E$. If Bob's first roll is an even number n_e, his expected sum is simply n_e. Therefore, $E = \frac{1}{6}((E+1)+(E+3)+(E+5))+\frac{1}{6}(2+4+6)$. Solving, we find $E = \boxed{7}$.

TR 6: The rhombus $PQRS$ has side length 3. The point X lies on segment PR such that line QX is perpendicular to line PS. Given $QX = 2$, the area of $PQRS$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{121}$

Solution:



Let C denote the center of the rhombus and let Y be the intersection of PS and QX. Note that $\triangle QCX \sim \triangle PYX \sim \triangle PCS \cong \triangle PCQ$, and that the ratio of the side lengths of $\triangle QCX$ to those of $\triangle PCQ$ is $\frac{QX}{PQ} = \frac{2}{3}$. Let $PC = a$ and $QC = b$. By similar triangles, we have $CX = \frac{2}{3}b$. From the Pythagorean Theorem on triangles QCX and PCQ, we have

$$b^2 + \left(\frac{2}{3}b\right)^2 = 4$$
$$a^2 + b^2 = 9.$$

Solving, we find that $(a, b) = \left(\frac{9}{\sqrt{13}}, \frac{6}{\sqrt{13}}\right)$. So, the area of the rhombus is

$$2ab = \frac{108}{13} \implies 108 + 13 = \boxed{121}.$$

TR 7: Let n be the largest integer such that $20!^{25!} + 25!^{20!}$ is divisible by 2025^n. The value of n can be expressed in the form $a \cdot b!$, where a and b are positive integers and b is maximized. Find $a + b$.

Answer: $\boxed{69}$

Solution: As usual, for a prime p and a positive integer n, let $v_p(n)$ denote the exponent of p in the prime factorization of n. First, we find the value of $v_3(20!^{25!} + 25!^{20!})$. Note that $v_3(20!^{25!}) = v_3(20!) \cdot 25! = 8 \cdot 25!$ and $v_3(25!^{20!}) = v_3(25!) \cdot 20! = 10 \cdot 20!$. Since $8 \cdot 25! > 10 \cdot 20!$, we have $v_3(20!^{25!} + 25!^{20!}) = v_3(25!) \cdot 20! = 10 \cdot 20!$. Using an analogous approach, we find that $v_5(20!^{25!} + 25!^{20!}) = 6 \cdot 20!$. Recall that $2025 = 3^4 \cdot 5^2$. Therefore,

$$\begin{aligned} n &= \min\left\{\left\lfloor \frac{10 \cdot 20!}{4} \right\rfloor, \left\lfloor \frac{6 \cdot 20!}{2} \right\rfloor\right\} \\ &= \min\{50 \cdot 19!, 3 \cdot 20!\} \\ &= 50 \cdot 19!. \end{aligned}$$

Since 20 does not divide 50, we cannot express $50 \cdot 19!$ as the product of a positive integer with a factorial greater than 19!. The answer is $50 + 19 = \boxed{69}$.

TR 8: Adam has a fair coin with a 2 written on one side and a 3 written on the other. What is the expected number of times Adam needs to flip the coin for the sum of all his coin flips to be a positive multiple of 36?

Answer: $\boxed{36}$

Solution: Let e_t denote the expected number of additional flips Adam needs for his sum to be a multiple of 36 when his current sum is congruent to t modulo 36, where at least one more flip is required. Let e'_t denote the same quantity without this requirement, so $e'_0 = 0$ and $e'_t = e_t$ for $t \neq 0$. Our answer is e_0. For each $0 \leq t \leq 35$, we have

$$e_t = \frac{1}{2}e'_{t+2} + \frac{1}{2}e'_{t+3} + 1,$$

where the subscripts are taken modulo 36. Since $t \mapsto t+2$ and $t \mapsto t+3$ are both bijections modulo 36, adding these 36 equations gives

$$\sum_{t=0}^{35} e_t = \sum_{t=0}^{35} e'_t + 36.$$

Since $e'_t = e_t$ for $t \neq 0$, this simplifies to $e_0 = e'_0 + 36 = \boxed{36}$.

TR 9: Pairwise distinct integers a, b, c, and d satisfy the system of equations

$$\begin{aligned} ab &= cd \\ a + d &= b + c \\ \frac{1}{a} + \frac{1}{b} - \frac{1}{5} &= \frac{1}{c} + \frac{1}{d} + \frac{1}{5}. \end{aligned}$$

What is the minimum possible value of $a^2 + b^2 + c^2 + d^2$?

Answer: $\boxed{832}$

Solution: Rearrange the second equation as $a - b = c - d$ and square it to obtain $a^2 - 2ab + b^2 = c^2 - 2cd + d^2$. Adding $4ab = 4cd$ to this equation, we obtain $(a+b)^2 = (c+d)^2$, or

$a+b=\pm(c+d)$. For the sake of contradiction, suppose $a+b=c+d$. Because $ab=cd$, from Vieta's formulas, we must have that $\{a,b\}=\{c,d\}$, which violates the condition that the four variables be pairwise distinct. Thus, we must have $a+b=-c-d$. This implies that $a+d=-b-c=b+c$, so $b+c=a+d=0$. Hence, (a,b,c,d) is of the form $(u,v,-v,-u)$, where u and v are distinct nonzero integers. Any quadruple $(a,b,c,d)=(u,v,-v,-u)$ satisfies the first two equations in the problem statement, so it suffices to determine the minimum possible value of $2(u^2+v^2)=a^2+b^2+c^2+d^2$ over the ordered pairs (u,v) satisfying the equation

$$\frac{1}{u}+\frac{1}{v}-\frac{1}{5}=-\frac{1}{u}-\frac{1}{v}+\frac{1}{5}.$$

Clearing the fractions and using Simon's Favorite Factoring Trick, we have $(u-5)(v-5)=25$. Keeping in mind that $u,v\neq 0$ and $u\neq v$, the only solutions to this equation in the integers are $(u,v)=(30,6),(-20,4)$ and permutations. By inspection, the minimum possible value of $2(u^2+v^2)$ is $2((-20)^2+4^2)=\boxed{832}$.

TR 10: Anna has a three-term arithmetic sequence of integers. She divides each term of her sequence by a positive integer $n>1$, and tells Bob that the three resulting remainders are 20, 52, and R, in some order. For how many values of R is it possible for Bob to uniquely determine n?

Answer: $\boxed{54}$

Solution: Let Anna's sequence be a, b, c, and let x, y, and z be the remainders when a, b, and c are divided by n, respectively. Since $b-a=c-b$, n divides $x-2y+z$. If $x-2y+z=0$, then n can be any integer greater than x, y, and z (for example, Anna's sequence could be x, y, z), so Bob cannot determine n. Otherwise, since $0\leq x,y,z<n$, we have $0<|x-2y+z|<2n$, so $n=|x-2y+z|$. If $n=x-2y+z$, then the conditions $n>x$, $n>y$, and $n>z$ become

$$z>2y,\quad x+z>3y,\quad \text{and}\quad x>2y,$$

which hold if and only if $x,z>2y$. If $n=2y-x-z$, then they become

$$2y>2x+z,\quad y>x+z,\quad \text{and}\quad 2y>x+2z,$$

which hold if and only if $y>x+z$. Conversely, if either condition holds, then the sequence x, y, $2y-x$ gives the remainders x, y, z, since $2y-x\equiv z \pmod{n}$. Let $u\leq v\leq w$ be 20, 52, and R in increasing order. The first condition can only hold with $y=u$, giving $n=v+w-2u$ when $v,w>2u$, and the second can only hold with $y=w$, giving $n=2w-u-v$ when $w>u+v$. If both hold, then Bob cannot determine n, since $v+w-2u=2w-u-v$ would mean that u, v, w form an arithmetic sequence. Also, if u, v, w form an arithmetic sequence, then $w=2v-u$, and both conditions are equivalent to $v>2u$. So, Bob can determine n if and only if exactly one of the conditions $v,w>2u$ and $w>u+v$ holds. If $R<20$, then $52>R+20$, so we need $2R\geq 20$, giving the 10 values $10\leq R\leq 19$. If $20\leq R<52$, then $52>2\cdot 20$, so we need exactly one of $R>40$ and $52>R+20$ to hold, giving the $12+11=23$ values $20\leq R\leq 31$ and $41\leq R\leq 51$. If $R\geq 52$, then $52>40$ and $R>40$, so we need $R\leq 72$, giving the 21 values $52\leq R\leq 72$. In conclusion, the answer is $10+23+21=\boxed{54}$.

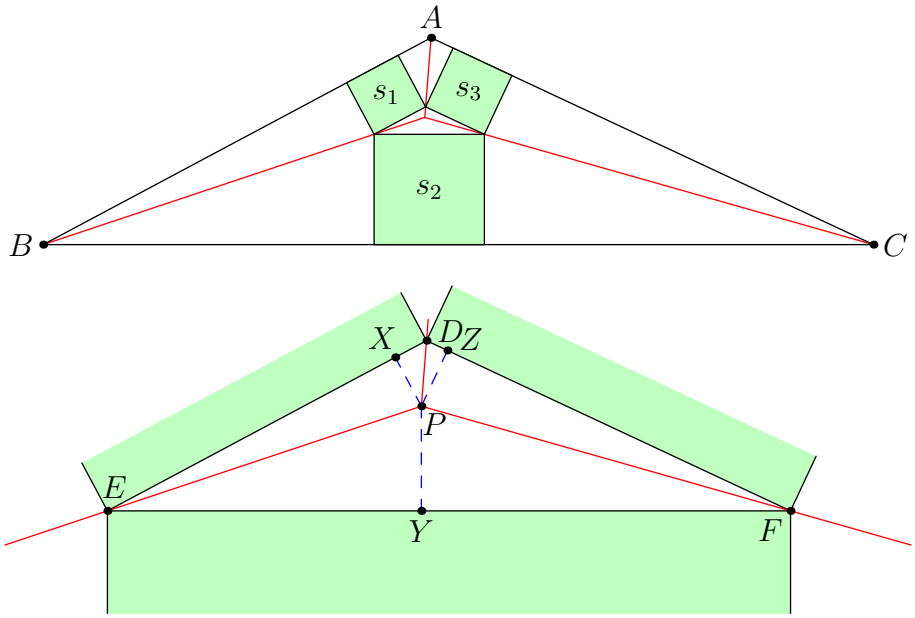
TR 11: Squares s_1, s_2, and s_3 with side lengths 9, 17, and 10, respectively, lie inside of $\triangle ABC$ such that:

- s_1 has a side that lies on $\overline{AB}$, s_2 has a side that lies on $\overline{BC}$, and s_3 has a side that lies on $\overline{CA}$;
- each square shares exactly one vertex with each of the other two squares.

Find the perimeter of $\triangle ABC$.

Answer: $\boxed{271}$

Solution:



Let D be the shared vertex of s_1 and s_3, let E be the shared vertex of s_1 and s_2, and let F be the shared vertex of s_2 and s_3. Since $AB \parallel ED$, $EF \parallel BC$, and $FD \parallel CA$, $\triangle ABC$ is the image of $\triangle DEF$ under a homothety with scale factor k. Therefore, lines AD, BE, and CF are concurrent and meet at the center of homothety, which we call P. Let X, Y, and Z be the projections of P onto DE, EF, and FD, respectively. By homothety, we have

$$\frac{PX+9}{PX} = \frac{PY+17}{PY} = \frac{PZ+10}{PZ} = k.$$

This implies that $PX : PY : PZ = 9 : 17 : 10$, so let $PX = 9h$, $PY = 17h$, and $PZ = 10h$ for some positive real h. Note that $k = 1 + \frac{1}{h}$. By Heron's formula, $[DEF] = 36$. However, we also have

$$\begin{aligned}[DEF] &= [DPE] + [EPF] + [FPD] \\ &= \frac{1}{2}(9 \cdot 9h + 17 \cdot 17h + 10 \cdot 10h) \\ &= 235h.\end{aligned}$$

Thus, $h = \frac{36}{235}$ and $k = 1 + \frac{235}{36} = \frac{271}{36}$. Since the perimeter of $\triangle DEF$ is 36, the perimeter of $\triangle ABC$ is $36k = \boxed{271}$.

TR 12: Let $a_k = (4 + 3\sqrt{2})^k$ for all nonnegative integers k. The value of

$$\sum_{k=0}^{\infty} \frac{\lfloor a_k \rfloor}{10^k}$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{659}$

Solution: Note that $(4+3\sqrt{2})^k + (4-3\sqrt{2})^k$ is an integer for every nonnegative integer k, and $-1 < 4-3\sqrt{2} < 0$. If $k > 0$ is even, we have $0 < (4-3\sqrt{2})^k < 1$. This means that $(4+3\sqrt{2})^k$ is less than the integer $(4+3\sqrt{2})^k + (4-3\sqrt{2})^k$ while being within less than 1 of it. If k is odd, $-1 < (4-3\sqrt{2})^k < 0$. Then, $(4+3\sqrt{2})^k$ is greater than the integer $(4+3\sqrt{2})^k + (4-3\sqrt{2})^k$ while being within less than 1 of it. Therefore,

$$\lfloor a_k \rfloor = \begin{cases} 1 & \text{if } k = 0 \\ (4+3\sqrt{2})^k + (4-3\sqrt{2})^k - 1 & \text{if } k > 0 \text{ is even} \\ (4+3\sqrt{2})^k + (4-3\sqrt{2})^k & \text{if } k > 0 \text{ is odd.} \end{cases}$$

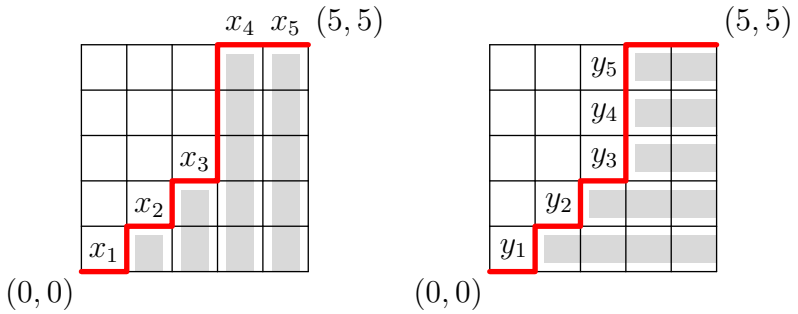
So,

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{\lfloor a_k \rfloor}{10^k} &= 1 + \sum_{k=1}^{\infty} \frac{(4+3\sqrt{2})^k}{10^k} + \sum_{k=1}^{\infty} \frac{(4-3\sqrt{2})^k}{10^k} - \sum_{k=1}^{\infty} \frac{1}{10^{2k}} \\ &= 1 + \frac{\frac{4+3\sqrt{2}}{10}}{1-\frac{4+3\sqrt{2}}{10}} + \frac{\frac{4-3\sqrt{2}}{10}}{1-\frac{4-3\sqrt{2}}{10}} - \frac{\frac{1}{100}}{1-\frac{1}{100}} \\ &= 1 + \frac{4+3\sqrt{2}}{6-3\sqrt{2}} + \frac{4-3\sqrt{2}}{6+3\sqrt{2}} - \frac{1}{99} \\ &= 1 + \frac{14}{3} - \frac{1}{99} \\ &= \frac{560}{99} \implies 560 + 99 = \boxed{659}. \end{aligned}$$

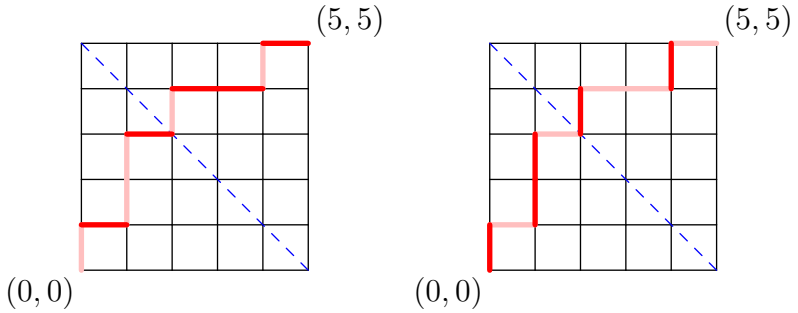
TR 13: Let S be the set of all ordered quintuples of integers $(x_1, x_2, x_3, x_4, x_5)$ satisfying $0 \le x_1 \le x_2 \le x_3 \le x_4 \le x_5 \le 5$. The *conjugate* of a quintuple in S is defined as $(y_1, y_2, y_3, y_4, y_5)$, where for each integer $1 \le i \le 5$, y_i is the number of indices $1 \le j \le 5$ satisfying $x_j \ge i$. The probability that a quintuple chosen uniformly at random from S is a permutation of its conjugate can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{71}$

Solution: Let P be the set of paths from $(0,0)$ to $(5,5)$ consisting of 10 unit steps, each up or right. For a step of such a path, define its *size* as its distance from the line $y = 0$ if the step is horizontal, and as its distance from the line $x = 5$ if the step is vertical. We map each quintuple $s = (x_1, x_2, x_3, x_4, x_5)$ in S to the path p in P whose i^{th} horizontal step has size x_i for each $1 \le i \le 5$, which gives a bijection between S and P. Let $s' = (y_1, y_2, y_3, y_4, y_5)$ be the conjugate of s. Since the ith vertical step of p lies to the left of exactly the horizontal steps with size at least i, its size is y_i.



Since $x_1 \le x_2 \le \cdots \le x_5$ and $y_1 \ge y_2 \ge \cdots \ge y_5$, s is a permutation of s' if and only if $x_i = y_{6-i}$ for all $1 \le i \le 5$. This means that the ith horizontal step of p has the same size as its $(6-i)^{\text{th}}$ vertical step, so these two steps are reflections of each other over the line $x+y=5$. So, s is a permutation of s' if and only if p is symmetric about the line $x+y=5$.



Such a path is determined by its first five steps, each of which can be up or right, so there are $2^5 = 32$ such paths. Since $|S| = |P| = \binom{10}{5} = 252$, the desired probability is

$$\frac{32}{252} = \frac{8}{63} \implies 8+63 = \boxed{71}.$$

TR 14: Find the number of ordered triples of positive integers (a,b,c) such that $a+b+c = 305$ and $ab+bc+ca$ is a multiple of 305.

Answer: $\boxed{60}$

Solution: Suppose $p \neq 3$ is a prime dividing both $a+b+c$ and $ab+bc+ca$. Consider the polynomial $f(x) = (x-a)(x-b)(x-c)$ modulo p. Expanding f, we have

$$\begin{aligned}(x-a)(x-b)(x-c) &\equiv x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc \pmod p \\ &\equiv x^3 - abc \pmod p.\end{aligned}$$

Therefore, the roots of f are the cube roots of abc modulo p. However, the roots of f are also a, b, and c. This gives us two possibilities. Either a root of f is 0 modulo p and f has a triple root at 0, or $3 \mid p-1$ and a, b, and c are the three distinct nonzero cube roots of abc modulo p. Since $3 \nmid 5-1$, we must have $a \equiv b \equiv c \equiv 0 \pmod 5$. Let $(a,b,c) = (5a_1, 5b_1, 5c_1)$ for positive integers a_1, b_1, and c_1 satisfying $a_1+b_1+c_1 = 61$. Since $0 < a_1, b_1, c_1 < 61$ and $3 \mid 61-1$, it follows that a_1, b_1, and c_1 are the three cube roots of a nonzero cubic residue modulo 61. Let g be a primitive root modulo 61; we partition the set $\{1, 2, \ldots, 60\}$ into the 20 sets $S_0, S_1, \ldots, S_{19}$, where for each integer $0 \le j \le 19$, we have $S_j \equiv \{g^j, g^{j+20}, g^{j+40}\} \pmod{61}$. Then, $\{a_1, b_1, c_1\}$ must be equal to some S_j. Note that the element sum of any S_j is a multiple of 61, since $g^{20} \not\equiv 1$ is a root of $x^3 - 1 = (x-1)(x^2+x+1)$, meaning

$g^j\left(1+g^{20}+g^{40}\right) \equiv 0 \pmod{61}$. Since this sum is between 6 and 177, it must be either 61 or 122. Recall that we must have $a_1+b_1+c_1 = 61$. Hence, we seek the number of j such that the element sum of S_j is 61. To do this, we show that there is in fact a bijection between the S_j that have an element sum of 61 and those that have one of 122. Suppose $S_j = \{u, v, w\}$ for some $0 \le j \le 9$, and consider S_{j+10}. Since $g^{30} \equiv -1 \pmod{61}$, we have

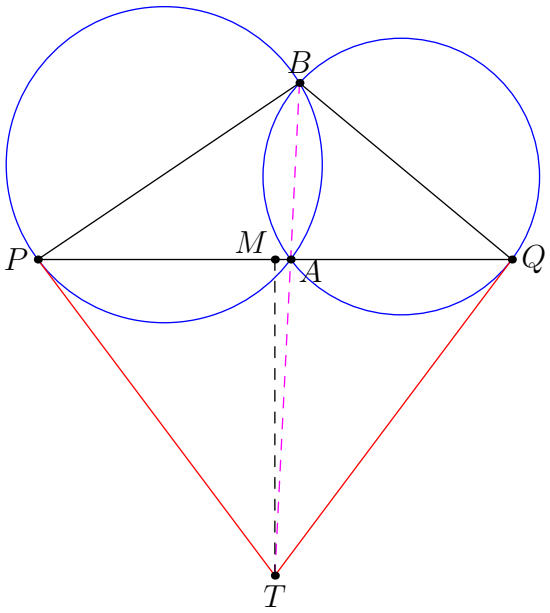
$$\begin{aligned} S_{j+10} &\equiv \{g^{j+10}, g^{j+30}, g^{j+50}\} \pmod{61} \\ &\equiv \{-g^{j+40}, -g^j, -g^{j+20}\} \pmod{61} \\ &\equiv \{-u, -v, -w\} \pmod{61}. \end{aligned}$$

Therefore, $S_{j+10} = \{61-u, 61-v, 61-w\}$. The element sum of S_j plus the element sum of S_{j+10} is 183, so exactly one of S_j and S_{j+10} has an element sum of 61 while the other has one of 122. To finish, there are $\frac{20}{2} = 10$ choices for $\{a_1, b_1, c_1\}$, and we can permute the three variables in $3! = 6$ ways. The answer is $6 \cdot 10 = \boxed{60}$.

TR 15: The circles ω_1 and ω_2 have radii 8 and 7, respectively, and intersect at points A and B. A line ℓ passing through A intersects ω_1 again at P and ω_2 again at Q, with $PQ = 24$. There exists a point T on line AB, with A between T and B, such that PT is tangent to ω_1 and QT is tangent to ω_2. The length of BT can be expressed as $\frac{m}{\sqrt{n}}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m+n$.

Answer: $\boxed{901}$

Solution:



Let M be the midpoint of PQ. From the power of a point at T, we have $TP^2 = TA \cdot TB = TQ^2$, so $TP = TQ$ and $TM \perp PQ$. From the tangent-chord angle theorem, $\angle PBA = \angle TPA$ and $\angle QBA = \angle TQA$. Since $\angle TPQ = \angle TQP$, line BA bisects $\angle PBQ$. Also, $\angle PAB$ and $\angle QAB$ are supplementary, so from the extended Law of Sines,

$$\frac{PB}{QB} = \frac{16\sin\angle PAB}{14\sin\angle QAB} = \frac{8}{7}.$$

From the angle bisector theorem, $\frac{PA}{AQ} = \frac{8}{7}$, so $PA = \frac{64}{5}$, $AQ = \frac{56}{5}$, and $AM = PA - PM = \frac{4}{5}$. Now,

$$TA \cdot AB = TA \cdot TB - TA^2 = TP^2 - TA^2 = PM^2 - AM^2 = 144 - \frac{16}{25} = \frac{3584}{25}.$$

Let $\theta = \angle TAP = \angle QAB$. Then, $TA\cos\theta = AM = \frac{4}{5}$, so $TA = \frac{4}{5\cos\theta}$ and $AB = \frac{896\cos\theta}{5}$. From the Law of Cosines on triangles PAB and QAB, we have

$$PB^2 = PA^2 + AB^2 + 2 \cdot PA \cdot AB\cos\theta \quad \text{and} \quad QB^2 = AQ^2 + AB^2 - 2 \cdot AQ \cdot AB\cos\theta.$$

Since $PB = 16\sin\theta$ and $QB = 14\sin\theta$, subtracting these gives

$$\begin{aligned} 60\sin^2\theta &= \frac{64^2 - 56^2}{25} + 48 \cdot AB\cos\theta \implies \\ 60 - 60\cos^2\theta &= \frac{192}{5} + \frac{43008\cos^2\theta}{5} \implies \\ \cos^2\theta &= \frac{1}{401}. \end{aligned}$$

So,

$$BT = TA + AB = \frac{4}{5\cos\theta} + \frac{896\cos\theta}{5} = \frac{4\sqrt{401}}{5} + \frac{896}{5\sqrt{401}} = \frac{500}{\sqrt{401}} \implies 500 + 401 = \boxed{901}.$$